

1. Ionising radiation

Ionising	Ionising radiation is radiation powerful enough to ionise particles; it was originally found to come from radioactive rocks, and it was not known whether it was beams of particles, or waves, or something else.
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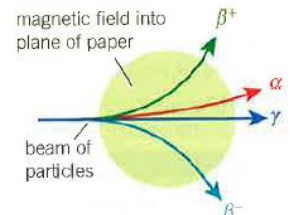
Three types of ionising radiation were identified, coming from radioactive materials; they were called types alpha beta and gamma. We now know about two sub-types of beta (beta+ and beta-)

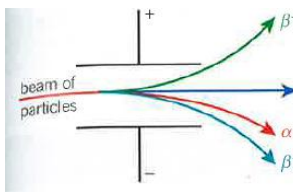
Radioactive	A material is radioactive if it emits ionising radiation. Some highly radioactive materials glow (emit visible light), but most do not!
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Types of ionising radiation	Alpha: highly ionising, weakly penetrating. Alpha particle are helium nuclei
	Beta-: medium ionising, medium penetrating; blocked by aluminium. Beta- particles are electrons
	Beta+: like beta- but positrons; these are antimatter and annihilate immediately on contact with an electron, so are not detected in normal radioactivity
	Gamma: weakly ionising, strongly penetrating. Electromagnetic waves.
	Neutron: neutral particles, so not ionising radiation! Included here because some radioisotopes emit neutrons

GM tube and counter	A Geiger-muller tube is used to detect ionising radiation. A high voltage is applied to the tube, and a spark (short burst of current) happens when ionisation occurs in the tube. A counter is used to count these sparks, or a ratemeter , which works out the count rate.
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Count rate	Number of counts per unit time detected by a GM tube (s^{-1})
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Use of magnetic field Moving, charged particles will be deflected by a magnetic field as in Chapter 23	 Particls will follow circular paths if they stay within the field
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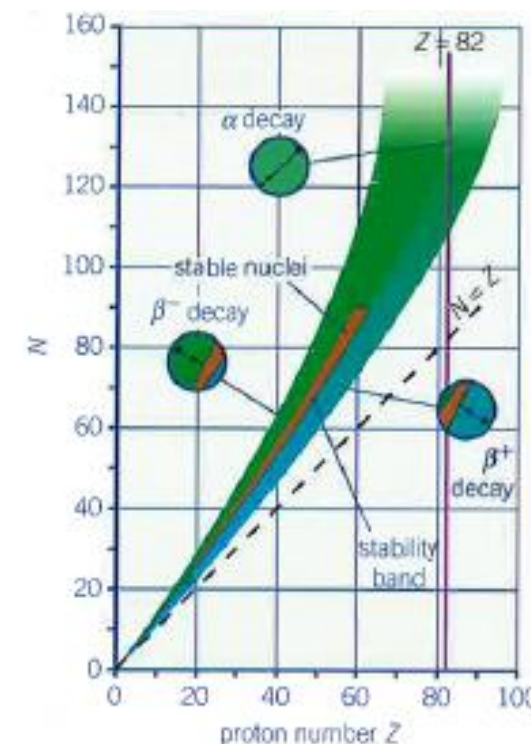
Use of electric field Charged particles will be deflected by an electric field as in Chapter 22	
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Background count	Radiation detected in the local environment, from radioactive rocks, gases, or cosmic rays. Very low levels
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Isotope notation	Isotopes are often written in the form Sr-90 ('strontium 90') rather than ${}^{90}_{38}\text{Sr}$
Radioisotope	A radioactive isotope: eg carbon-14 is a radioisotope, whereas carbon-12 is not
Contaminated	Radioactive material, eg dust, is on an object/person
Irradiated	Ionising radiation affects an object or person. Strawberries are often irradiated to kill microorganisms; the fruit is not contaminated , it does not become radioactive

2. Nuclear decay equations

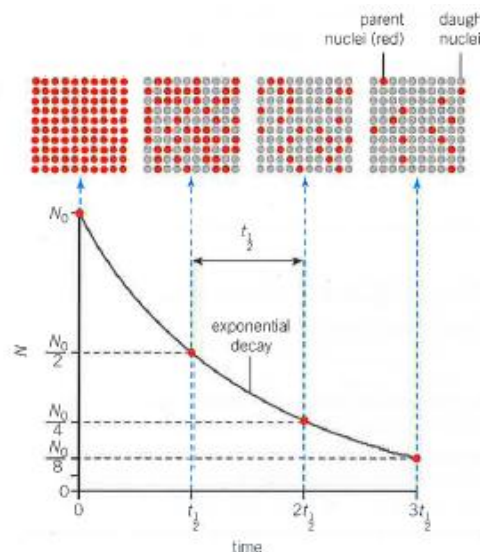
Nuclear decay	A random, spontaneous breakdown of nuclei, releasing ionising radiation
Parent nucleus	A nucleus before radioactive decay
Daughter nucleus	A nucleus after radioactive decay
Decay chain	A series a radioactive decays, where a parent nucleus gives rise to a radioactive daughter nucleus, which gives rise to another, continuing until a stable daughter nucleus occurs. For example: $\text{Ra-226} \rightarrow \text{Rn-222} \rightarrow \text{Po-218} \rightarrow \text{Pb-214} \rightarrow \text{Bi-214} \rightarrow \text{Po-214} \rightarrow \text{Pb-210} \rightarrow \text{Bi-210} \rightarrow \text{Po-210} \rightarrow \text{Pb-206}$
Stability curve or stability band	All known nuclei are plotted on an N-Z plot (N=neutron number, Z=proton number), and the stable nuclei are all found to lie in a band



In this topic N is used both for number of neutrons (eg in the N-Z plot) and number of radioactive particles (eg in the exponential decay equations). Take care!

3. Half-life and activity

Random	Nuclear decay is random: it is impossible to predict when the next decay of a nucleus in a sample will occur, or when a particular nucleus will decay.
Spontaneous	Nuclear decay occurs spontaneously, ie without a cause; it cannot be affected by external factors such as pressure or temperature
Half life, $t_{1/2}$	The time for the number of radioactive particles in a sample to halve; also The time for the <i>activity</i> of a sample to halve
Activity (A, unit Bq)	The number of decays per unit time in a sample $A = \frac{\Delta N}{\Delta t}$
Becquerel (Bq)	Unit of activity; number of decays per second in a sample. [Although this is s^{-1} , as for frequency, it is not interchangeable with Hz, number of cycles or waves per second]
Decay constant λ	The probability of decay of an individual nucleus, per unit time λ , where $A = \lambda N$ where N is the number of undecayed nuclei. Units s^{-1} .



4. Radioactive decay calculations

Exponential decay	The activity of a sample is proportional to the number of radioactive particles, N, remaining in a sample: $\frac{\Delta N}{\Delta t} = -\lambda N$ The solution is $N = N_0 e^{-\lambda t}$ Where N_0 is the original number of particles in the sample. The decay curve shown above matches this equation
Experimental data	No experimental data will match this curve exactly because of the random nature of radioactive decay. The larger N is, the better the agreement.

5. Modelling radioactive decay

Iterative process	The exponential decay curve can also be produced by an iterative process – the number of particles is calculated over small increments, usually in a spreadsheet
Time intervals	The smaller the time interval, the better the match with experimental data

6. Radioactive dating

	A technique to determine the age of objects, using isotopes with known half-lives
Carbon-14 and once-living objects (eg wood, bone)	Carbon-14 is a radioisotope created in the atmosphere by cosmic rays and absorbed into living things. It decays with a half-life of 5700 years. Assuming an organism had the same ratio of C-12 to C-14 in it as the atmosphere at the time of death, the time since death can be worked out by finding this ratio at the present time. The Turin Shroud was famously dated to around 1300AD using this technique
Uranium and rocks	Rocks can be dated using a similar technique, but isotopes with much longer half-lives are required (eg Rb-87 found in some rocks, half-life 49 billion years)

